

Design of a Thermal Comfort Liner for Transtibial Prosthesis Users to Manage Residual Limb Temperature

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Introduction

Individuals with lower-limb amputation often experience thermal discomfort with their silicone liners after prolonged use.

- Over 53% of lower limb amputees report discomfort due to heat and perspiration.
- 66% report that sweating impacts their daily activities.
- Bacterial infection-related complications occur in 20-41% of amputees because of friction between the limb and liner.

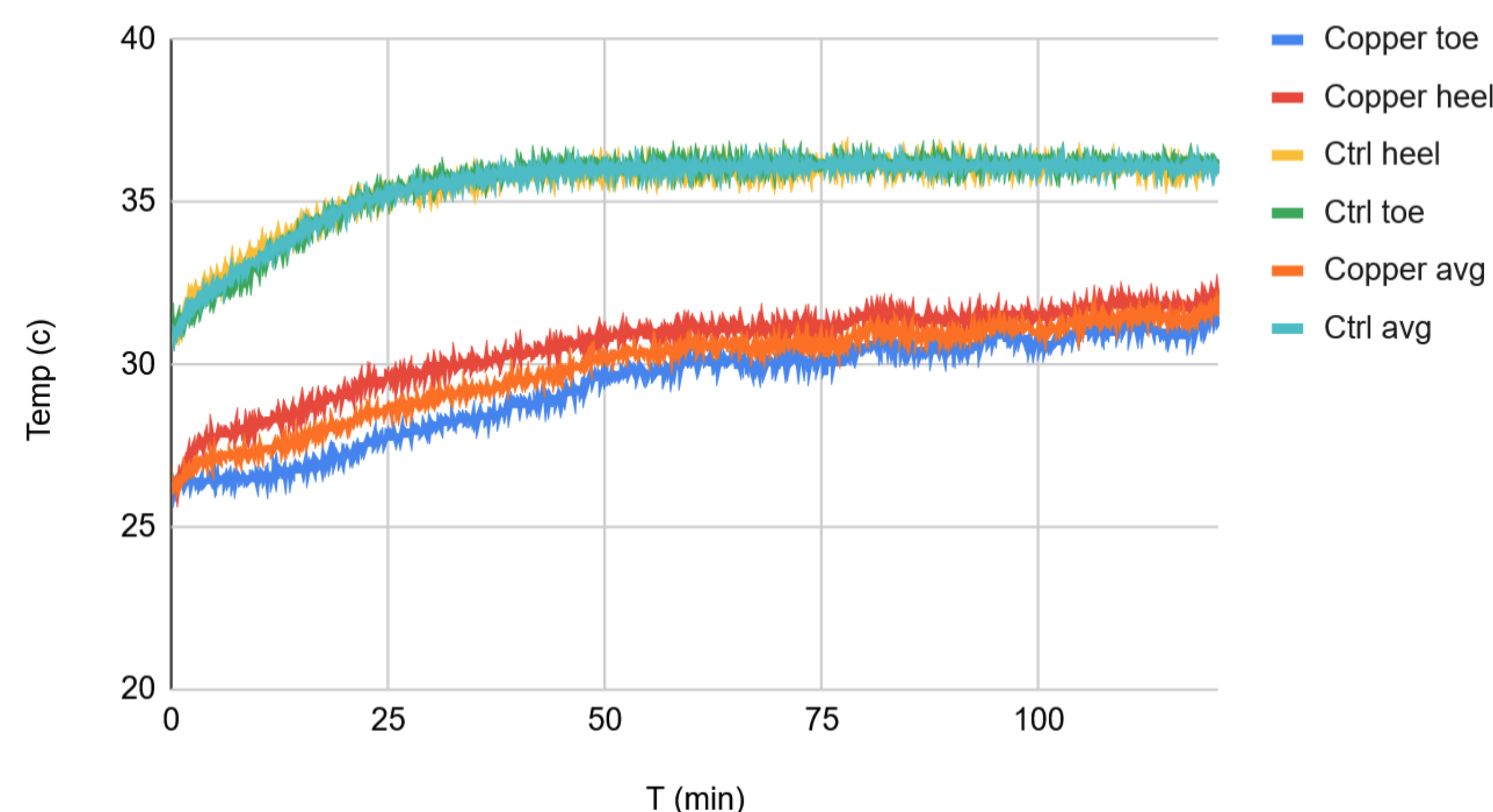
The objective of this project is to develop a liner that lowers the temperature of the residual limb while maintaining ease of donning and doffing.

Background

Foot-Insole Model

- Insole experimentation ran concurrently with liner design development, with findings directly informing liner material selection and allowing evaluation of heat transfer performance.
- Copper selected as a passive, heat-conductive method with a comparable thermal performance without the weight or power demands of an active system (Fig. A).

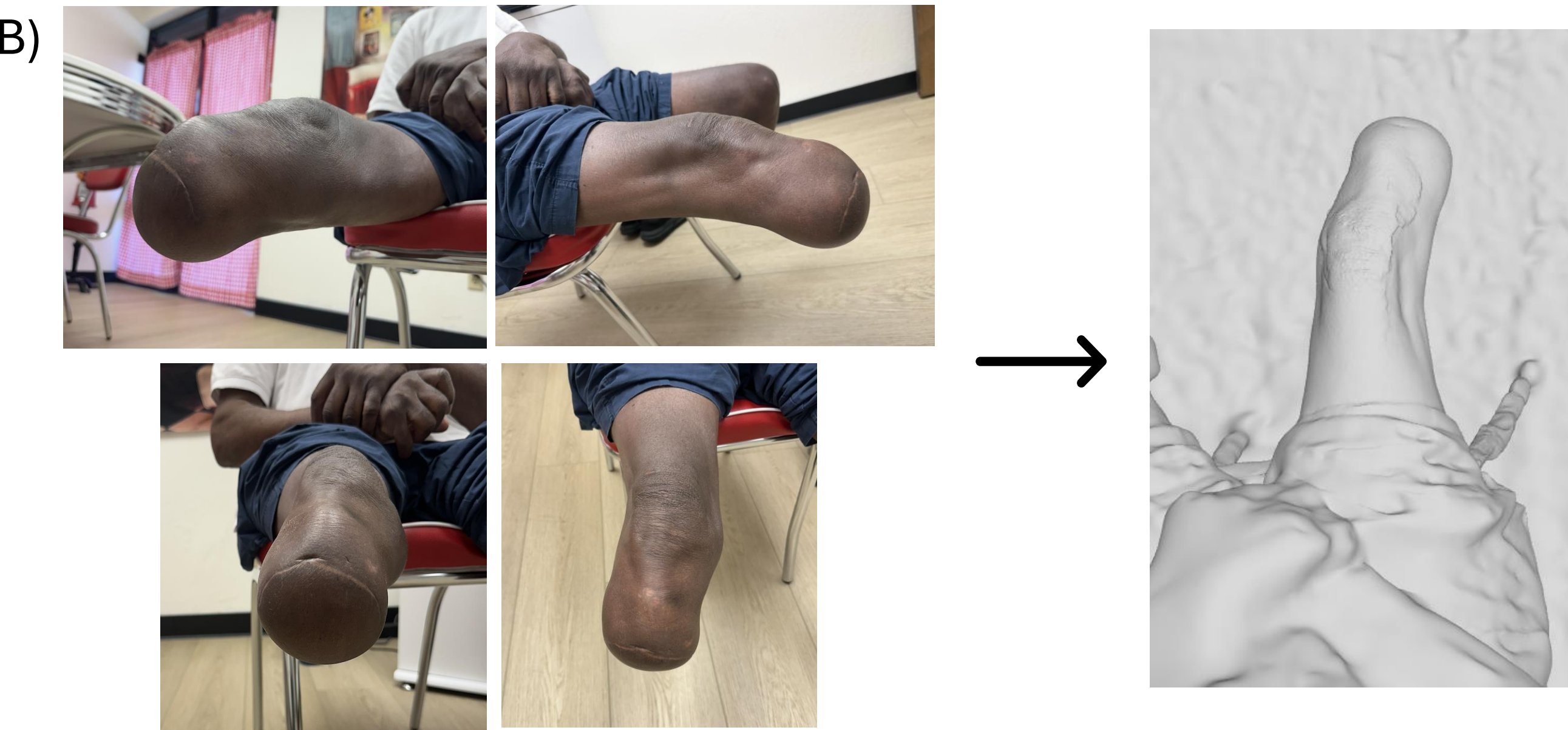
A) Copper Insole Temperatures



Methods

Residual Limb Scanning

- Captured 3D geometry of participant's residual limb using Kiri Engine, a smartphone-based photogrammetry/LiDAR scanning app.
- 150 circumferential images were taken of the participant's extended limb in a seated position (Fig. B).
- Cleaned and repaired the mesh in Blender to prepare for printing and flattening.



Liner Scanning

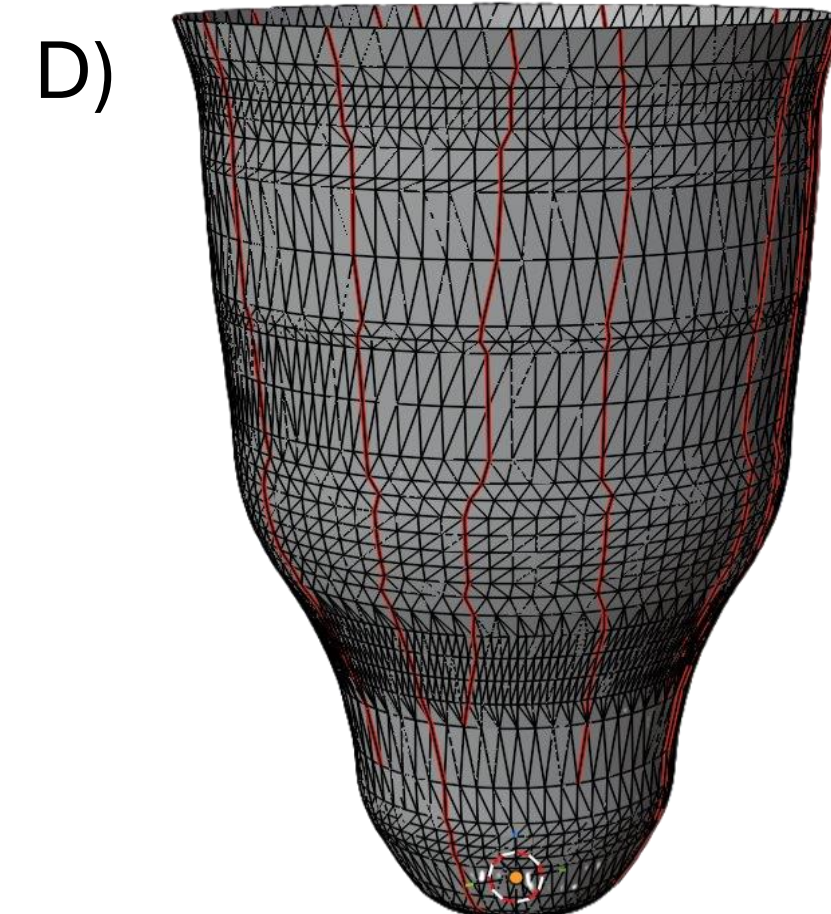
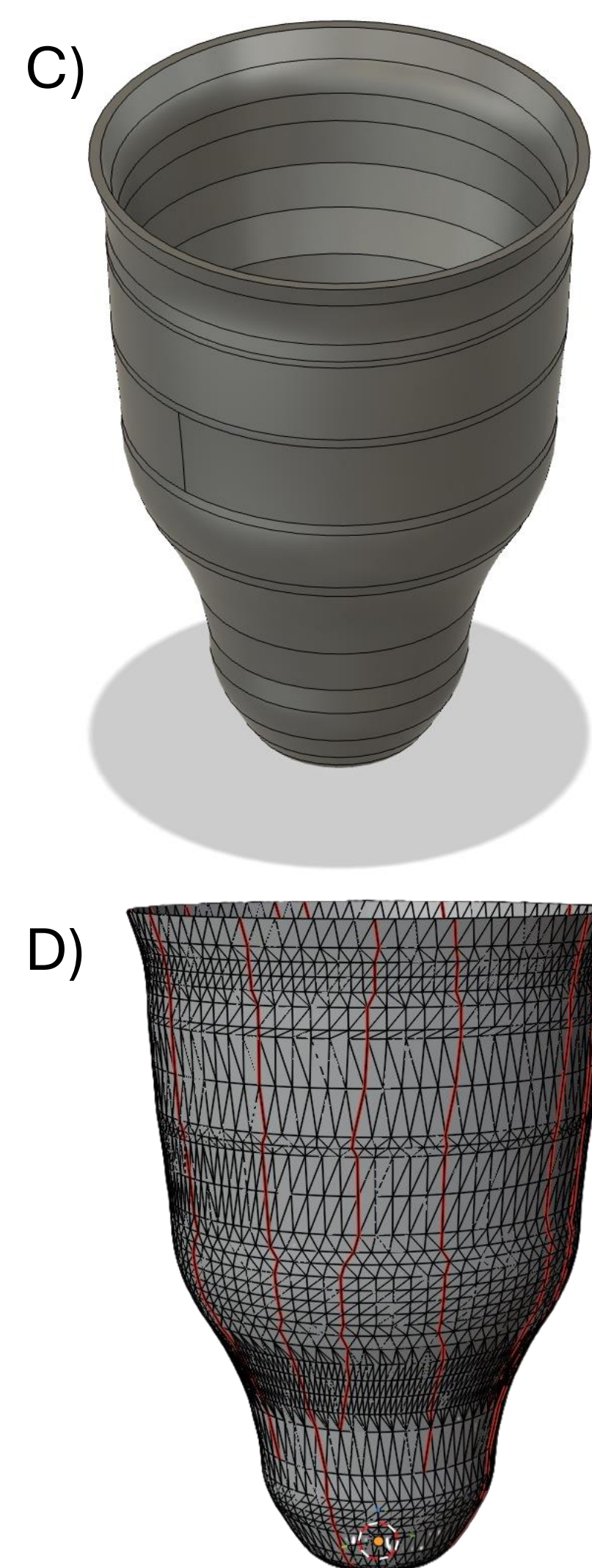
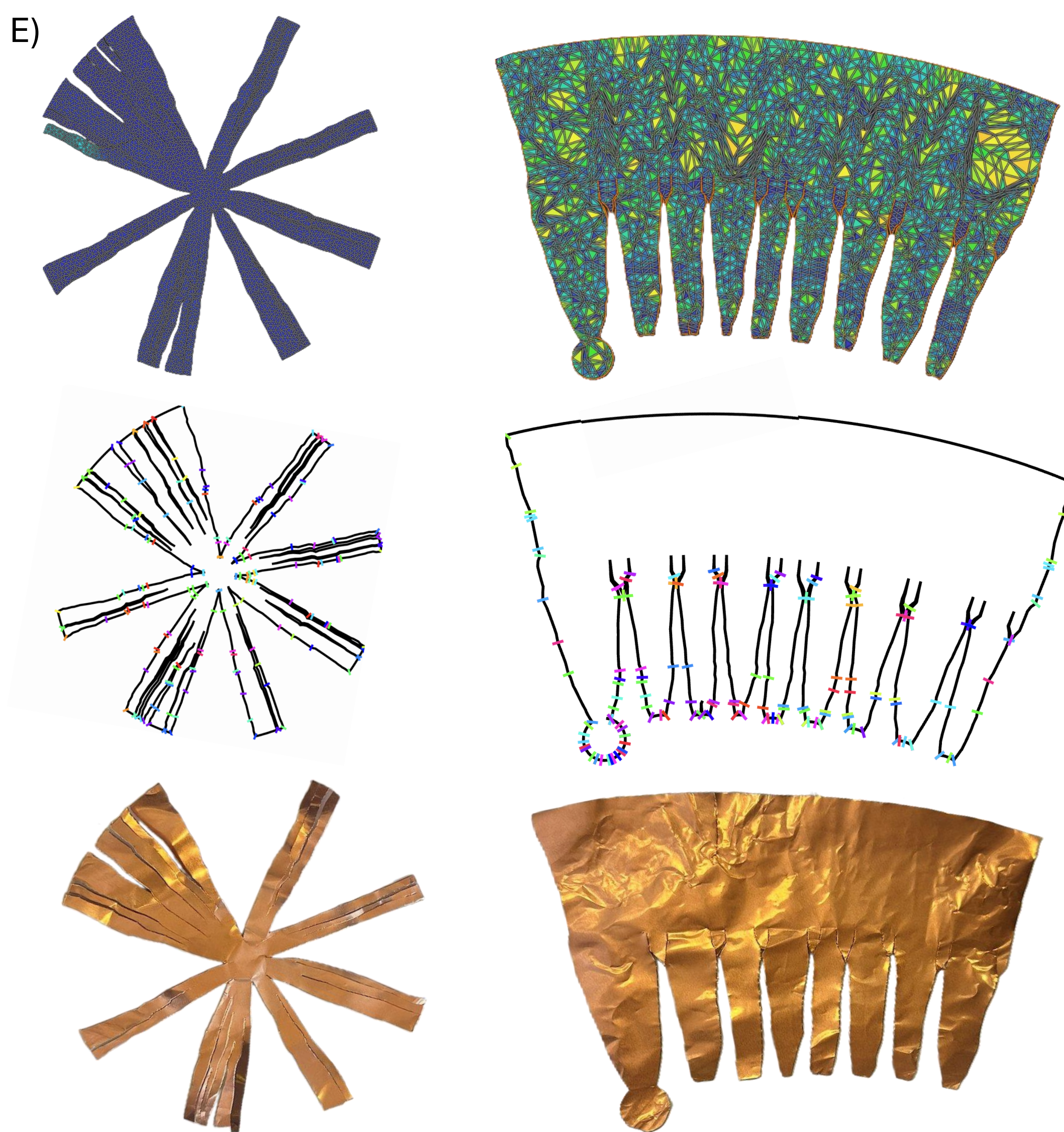
- Scanned the liner using the same protocol as the residual limb, with post-processing modified to account for its simpler, more uniform geometry.
- Sliced the liner mesh into ¼-inch height increments and fit each slice with a circle of median circumference.
- Lofted the stacked circles to reconstruct a cylindrically symmetrical liner model (Fig. C).

Seam Placement

- Implemented Thomas Kole's "Seams to Sewing Pattern" add-on in Blender to convert a seamed, UV-unwrapped 3D mesh into a 2D sewing pattern (Fig. D, E).
- Constrained seams to keep the pattern on a single sheet of material and unfold on a single cut.

Curvature-Based Seam Placement

- Computes discrete Gaussian curvature at every vertex using the angle-defect method marking cone-like features that need seams.
- Used curvature-weighted farthest-point sampling to distribute cut targets across the surface, preventing seams from clustering at a single dominant feature.
- Peaks connected using a curvature-weighted graph that favored valleys and avoided existing seams.
- Iterates until there are no peaks or unable to cut from a single sheet.



Residual Limb Model

- 3D printed the participant's residual limb scan in PETG and coated in silicone and epoxy to create a physical model for experimentation and testing (Fig. F).
- Fabricated a 1:3.5 scale model of the residual limb to prototype and validate design iterations before committing to full-scale fabrication (Fig. G).



Next Steps

Improve Residual Limb Model

- Print the residual limb model in TPU to better simulate the flexibility of skin and soft tissue.
- Simulate blood flow through the limb using a pump circulating heated water.
- Incorporate sweating into the model to more closely replicate real-world thermal conditions.

Transition from Insole to Liner

- Adapt a similar cooling approach to the one used in the cooling insole design.
- Further explore designs to optimize for ease of donning and doffing the liner.

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